



# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

**SENIOR CERTIFICATE EXAMINATIONS/  
NATIONAL SENIOR CERTIFICATE EXAMINATIONS/  
*SENIORSERTIFIKAAT-EKSAMEN/  
NASIONALE SENIORSERTIFIKAAT-EKSAMEN***

**MATHEMATICS P2/*WISKUNDE V2***

**MAY/JUNE 2026/*MEI/JUNIE 2026***

**MARKING GUIDELINES/*NASIENRIGLYNE***

**MARKS/*PUNTE*: 150**

**These marking guidelines consist of 24 pages./  
*Hierdie nasienriglyne bestaan uit 24 bladsye.***

**NOTE:**

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out an attempt of a question and not redone the question, mark the crossed-out version.
- Consistent accuracy applies in ALL aspects of the Marking Guidelines. Stop marking at the second calculation error.
- Assuming answers/values in order to solve a problem is NOT acceptable.

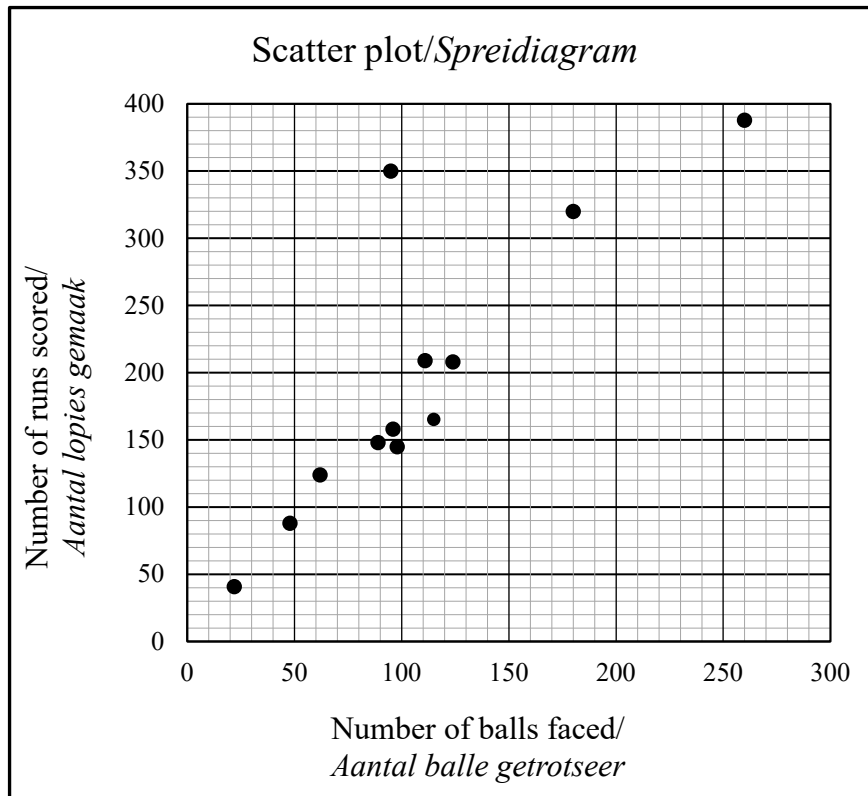
**LET WEL:**

- *As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.*
- *As 'n kandidaat 'n antwoord van 'n vraag doodtrek en nie oordoen nie, sien die doodgetrekte poging na.*
- *Volgehoue akkuraatheid word in ALLE aspekte van die Nasienriglyne toegepas. Hou op nasien by die tweede berekeningsfout.*
- *Aanvaar van antwoorde/waardes om 'n probleem op te los, word NIE toegelaat nie.*

| <b>GEOMETRY • MEETKUNDE</b> |                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>S</b>                    | <b>A mark for a correct statement<br/>(A statement mark is independent of a reason)</b>                       |
|                             | <b>'n Punt vir 'n korrekte bewering<br/>( 'n Punt vir 'n bewering is onafhanklik van die rede)</b>            |
| <b>R</b>                    | <b>A mark for the correct reason<br/>(A reason mark may only be awarded if the statement is correct)</b>      |
|                             | <b>'n Punt vir 'n korrekte rede<br/>( 'n Punt word slegs vir die rede toegeken as die bewering korrek is)</b> |
| <b>S/R</b>                  | <b>Award a mark if statement AND reason are both correct</b>                                                  |
|                             | <b>Ken 'n punt toe as die bewering EN rede beide korrek is</b>                                                |

**QUESTION/VRAAG 1**

|                                                                          |     |     |    |    |     |     |     |     |     |     |     |     |
|--------------------------------------------------------------------------|-----|-----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|
| <b>NUMBER OF BALLS<br/>FACED (x)/<br/>GETAL BALLE<br/>GETROTSEER (x)</b> | 62  | 111 | 22 | 48 | 124 | 180 | 89  | 96  | 95  | 260 | 98  | 115 |
| <b>NUMBER OF RUNS<br/>SCORED (y)/<br/>GETAL LOPIES GEMAAK<br/>(y)</b>    | 124 | 209 | 41 | 88 | 208 | 320 | 148 | 158 | 350 | 388 | 145 | 165 |

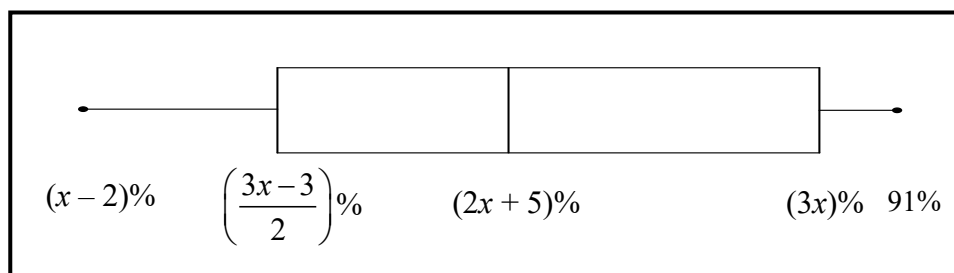


|             |                                                                                                                                                                                                               |                                                                                              |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1.1         | $a = 39,47$<br>$b = 1,44$<br>$\hat{y} = 39,47 + 1,44x$                                                                                                                                                        | ✓ $a = 39,47$<br>✓ $b = 1,44$<br>✓ equation<br>(3)                                           |
| 1.2         | $y = 39,47 + 1,44(150)$<br>$y = 255,47$<br><b>OR</b><br>$y = 255,28$ (calculator)                                                                                                                             | ✓ substitution<br>✓ answer<br>(2)<br><br>✓✓ answer<br>(2)                                    |
| 1.3         | $\bar{y} = 195,33$<br>$\sigma_y = 101,84$<br>One $\sigma$ above the $\bar{y}$ : $195,33 + 101,84$<br>$= 297,17$<br>No. of players = 3                                                                         | ✓ $\bar{y} = 195,33$<br>✓ $\sigma_y = 101,84$<br>✓ $\bar{y} + \sigma$<br><br>✓ answer<br>(4) |
| 1.4         | This player scored many more runs than the other players who faced a similar number of balls. / Hierdie speler het meer lopies as enige ander speler, wat dieselfde hoeveelheid balle getrotseer het, gemaak. | ✓ answer<br>(1)                                                                              |
| <b>[10]</b> |                                                                                                                                                                                                               |                                                                                              |

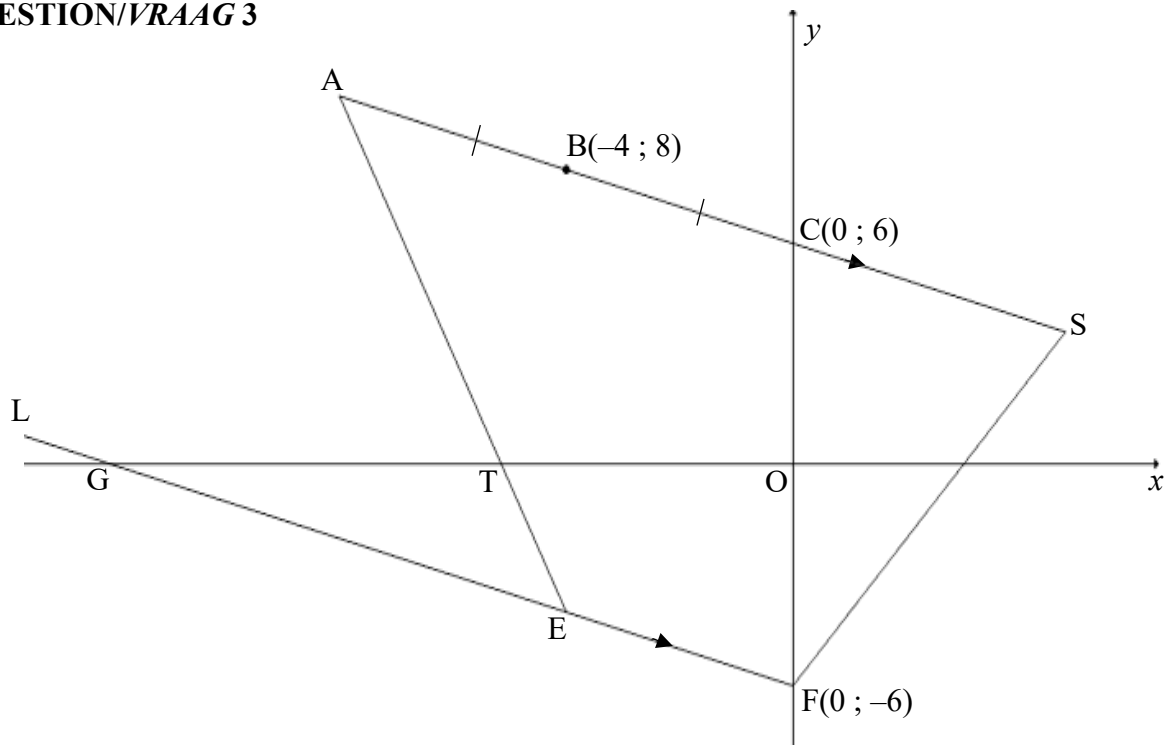
## QUESTION/VRAAG 2

| 2.1.1                                                              | $50 < x \leq 60$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ✓ $50 < x \leq 60$<br>(1)                                                                                       |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------|------------------|---|---|------------------|----|----|------------------|----|----|------------------|----|----|------------------|----|----|------------------|---|----|------------------|---|----|-------------------|---|----|--------------------------------------------------------------------------------------------------|
| 2.1.2                                                              | <table border="1"> <thead> <tr> <th>CLASS INTERVALS<br/>(AS A %)/<br/><i>KLASINTERVALLE</i><br/>(AS 'N %)</th><th>FREQUENCY/<br/><i>FREKWENSIE</i></th><th>CUMULATIVE<br/>FREQUENCY/<br/><i>KUMULATIEWE</i><br/><i>FREKWENSIE</i></th></tr> </thead> <tbody> <tr> <td><math>20 &lt; x \leq 30</math></td><td>8</td><td>8</td></tr> <tr> <td><math>30 &lt; x \leq 40</math></td><td>14</td><td>22</td></tr> <tr> <td><math>40 &lt; x \leq 50</math></td><td>16</td><td>38</td></tr> <tr> <td><math>50 &lt; x \leq 60</math></td><td>21</td><td>59</td></tr> <tr> <td><math>60 &lt; x \leq 70</math></td><td>16</td><td>75</td></tr> <tr> <td><math>70 &lt; x \leq 80</math></td><td>9</td><td>84</td></tr> <tr> <td><math>80 &lt; x \leq 90</math></td><td>6</td><td>90</td></tr> <tr> <td><math>90 &lt; x \leq 100</math></td><td>2</td><td>92</td></tr> </tbody> </table> | CLASS INTERVALS<br>(AS A %)/<br><i>KLASINTERVALLE</i><br>(AS 'N %)                                              | FREQUENCY/<br><i>FREKWENSIE</i> | CUMULATIVE<br>FREQUENCY/<br><i>KUMULATIEWE</i><br><i>FREKWENSIE</i> | $20 < x \leq 30$ | 8 | 8 | $30 < x \leq 40$ | 14 | 22 | $40 < x \leq 50$ | 16 | 38 | $50 < x \leq 60$ | 21 | 59 | $60 < x \leq 70$ | 16 | 75 | $70 < x \leq 80$ | 9 | 84 | $80 < x \leq 90$ | 6 | 90 | $90 < x \leq 100$ | 2 | 92 | ✓ first four correct values<br><br>✓ last four values<br><br><br><br><br><br><br><br><br><br>(2) |
| CLASS INTERVALS<br>(AS A %)/<br><i>KLASINTERVALLE</i><br>(AS 'N %) | FREQUENCY/<br><i>FREKWENSIE</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CUMULATIVE<br>FREQUENCY/<br><i>KUMULATIEWE</i><br><i>FREKWENSIE</i>                                             |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $20 < x \leq 30$                                                   | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8                                                                                                               |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $30 < x \leq 40$                                                   | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 22                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $40 < x \leq 50$                                                   | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 38                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $50 < x \leq 60$                                                   | 21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 59                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $60 < x \leq 70$                                                   | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 75                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $70 < x \leq 80$                                                   | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 84                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $80 < x \leq 90$                                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 90                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| $90 < x \leq 100$                                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 92                                                                                                              |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| 2.1.3                                                              | <p style="text-align: center;"><i>Ogive/Ogief</i></p> <p style="text-align: center;">Cumulative Frequency/<br/><i>Kumulatiewe frekwensie</i></p> <p style="text-align: center;">Marks obtained by learners (as a percentage)/<br/><i>Punte wat leerders behaal het (as 'n persentasie)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✓ Grounding at (20 ; 0)<br>✓ plotting cumulative frequency at upper boundaries<br>✓ smooth curve<br><br><br>(3) |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |
| 2.1.4                                                              | Median = 54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ answer<br>(1)                                                                                                 |                                 |                                                                     |                  |   |   |                  |    |    |                  |    |    |                  |    |    |                  |    |    |                  |   |    |                  |   |    |                   |   |    |                                                                                                  |

2.2



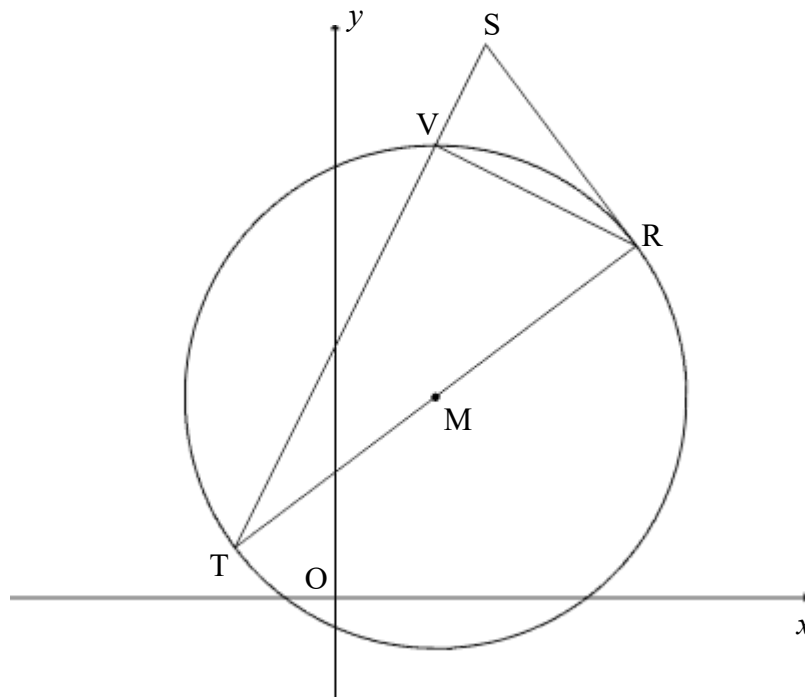
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| 2.2         | <p>IQR as a %: <math>3x - \frac{3x-3}{2} = 45</math></p> <p><math>6x - 3x + 3 = 90</math></p> <p><math>x = 29</math></p> <p>Min % = <math>(29 - 2)\%</math></p> <p><math>= 27\%</math></p> | <p>✓ <math>3x - \frac{3x-3}{2} = 45</math></p> <p>✓ value of <math>x</math></p> <p>✓ answer</p> <p>(3)</p> |
| <b>[10]</b> |                                                                                                                                                                                            |                                                                                                            |

**QUESTION/VRAAG 3**

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                             |
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| 3.1 | $\frac{x_A + 0}{2} = -4 \quad \frac{y_A + 6}{2} = 8$ $x_A = -8 \quad y_A = 10$ $A(-8; 10)$ <p><b>OR</b></p> $C \rightarrow B : (x; y) \rightarrow (x - 4; y + 2)$ $B \rightarrow A : (-4; 8) \rightarrow (-4 - 4; 8 + 2)$ $\therefore A(-8; 10)$ <p><b>OR</b></p> $m_{AS} = \frac{8 - 6}{-4 - 0}$ $m_{AS} = -\frac{1}{2}$ $y = -\frac{1}{2}x + 6$ $-\frac{1}{2}x + 6 = -\frac{7}{2}x - 18$ $-x + 12 = -7x - 36$ $6x = -48$ $x = -8$ $y = -\frac{1}{2}(-8) + 6$ $y = 10$ $A(-8; 10)$ | <p>✓ substitution into midpoint formula<br/>         ✓ <math>x_A</math> ✓ <math>y_A</math> (3)</p> <p>✓ translation: 4 left and 2 up from B<br/>         ✓ <math>x_A</math> ✓ <math>y_A</math> (3)</p> <p>✓ <math>-\frac{1}{2}x + 6 = -\frac{7}{2}x - 18</math></p> <p>✓ <math>x_A</math></p> <p>✓ <math>y_A</math> (3)</p> |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 3.2 | $m_{AS} = \frac{8-6}{-4-0}$ $m_{AS} = -\frac{1}{2}$ $m_{LF} = m_{AS} = -\frac{1}{2} \text{ (AS  LF)}$ $y = -\frac{1}{2}x - 6$                                                                                                                                                                                                                                                                                                                                                                             | ✓ substitution into gradient formula<br>✓ $m_{AS}$<br>✓ $c = -6$                                                            |
| 3.3 | $-\frac{1}{2}x - 6 = -\frac{7}{2}x - 18$ $-x - 12 = -7x - 36$ $6x = -24$ $x = -4$ $y = -\frac{1}{2}(-4) - 6$ $y = -4$ $E(-4; -4)$                                                                                                                                                                                                                                                                                                                                                                         | ✓ equating<br>✓ $x_E$<br>✓ $y_E$                                                                                            |
| 3.4 | $m_{AS} = -\frac{1}{2}$ $m_{FS} = \frac{\frac{18}{5} - (-6)}{\frac{24}{5} - 0} \quad \text{OR} \quad m_{FS} = \frac{-6 - \frac{18}{5}}{0 - \frac{24}{5}}$ $= 2$ $m_{AS} \times m_{FS} = -\frac{1}{2} \times 2$ $= -1$ $\therefore \hat{CSF} = 90^\circ$ <p>CF is a diameter [CF subtends a right <math>\angle</math>]<br/> / CF is 'n middellyn [omgekeerde <math>\angle</math> in halwe sirkel]<br/> <math>\therefore</math> centre O(0 ; 0) and <math>r = 6</math><br/> <math>x^2 + y^2 = 36</math></p> | ✓ $m_{FS}$<br>✓ $m_{AS} \times m_{FS} = -1$<br>✓ $\hat{CSF} = 90^\circ$ or CF is a diameter<br>✓ $r$<br>✓ $x^2 + y^2 = r^2$ |
| 3.5 | $\tan \hat{ATO} = m_{AE} = -\frac{7}{2}$ $\hat{ATO} = 105,95\dots^\circ$ $\tan \hat{LGT} = m_{LF} = -\frac{1}{2}$ $\hat{LGT} = 153,43\dots^\circ$ $\hat{AEG} = 153,43\dots^\circ - 105,95\dots^\circ$ $= 47,49^\circ$ <p><b>OR</b></p>                                                                                                                                                                                                                                                                    | ✓ $\tan \hat{ATO} = m_{AE}$ or<br>$\tan \hat{LGT} = m_{LF}$<br>✓ $\hat{ATO}$<br>✓ $\hat{LGT}$<br>✓ answer                   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                 |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | $\tan \hat{A}\hat{T}O = m_{AE} = -\frac{7}{2}$ $\hat{A}\hat{T}O = 105,95\dots^\circ$ <p>Inclination of AS produced = <math>153,43\dots^\circ</math></p> $\hat{T}\hat{A}S = 47,49^\circ \quad [\text{ext. } \angle \text{ s of } \Delta / \text{buite } \angle \text{ van } \Delta]$ $\hat{A}\hat{E}G = 47,49^\circ \quad [\text{alt } \angle \text{ s; } LF \parallel AS / \text{verw } \angle \text{ s; } LF \parallel AS]$ <p><b>OR</b></p> $G(-12; 0) \text{ and } T\left(-\frac{36}{7}; 0\right)$ $TG = \frac{48}{7} \text{ units}$ $GE = \sqrt{80} \text{ units}$ $TE = \frac{4\sqrt{53}}{7} \text{ units}$ $\left(\frac{48}{7}\right)^2 = (\sqrt{80})^2 + \left(\frac{4\sqrt{53}}{7}\right)^2 - 2(\sqrt{80})\left(\frac{4\sqrt{53}}{7}\right)\cos \hat{A}\hat{E}G$ $\hat{A}\hat{E}G = 47,49^\circ$ | <p>✓ <math>\tan \hat{A}\hat{T}O = m_{AE}</math></p> <p>✓ <math>\hat{A}\hat{T}O</math></p><br><p>✓ <math>\hat{T}\hat{A}S</math></p> <p>✓ answer</p> <p style="text-align: right;">(4)</p><br><p>✓ length of TG</p><br><p>✓ length of GE and TE</p><br><p>✓ substitution in cosine-rule</p> <p>✓ answer</p> <p style="text-align: right;">(4)</p> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>[18]</b>                                                                                                                                                                                                                                                                                                                                     |

**QUESTION/VRAAG 4**

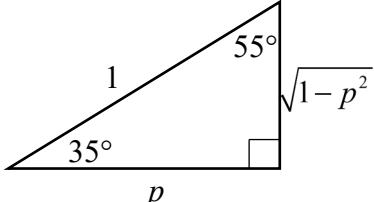
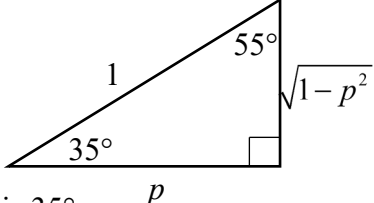
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| 4.1 | M(2 ; 4)                                                                                                                                                                                                                        | ✓ $x = 2$ ✓ $y = 4$<br>(2)                                                                                                    |
| 4.2 | $(x - 2)^2 + (2x + 5 - 4)^2 = 25$<br>$x^2 - 4x + 4 + 4x^2 + 4x + 1 - 25 = 0$<br>$5x^2 - 20 = 0$<br>$x^2 - 4 = 0$<br>$(x - 2)(x + 2) = 0$<br>$x = 2$ or $x = -2$<br><br>$y = 2(-2) + 5$<br>$y = 1$<br><br>$\therefore T(-2 ; 1)$ | ✓ substituting $y = 2x + 5$ into equation of circle<br>✓ standard form<br>✓ factors<br><br>✓ substituting $x = -2$<br><br>(4) |
| 4.3 | V(2 ; 9)                                                                                                                                                                                                                        | ✓ $x_V = 2$ ✓ $y_V = 9$<br>(2)                                                                                                |
| 4.4 | $m_{TR} = m_{TM} = \frac{4 - 1}{2 + 2}$<br>$= \frac{3}{4}$                                                                                                                                                                      | ✓ substitution into the gradient formula<br>✓ answer<br><br>(2)                                                               |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.5 | <p>R(6;7) [Translation <b>OR</b> midpoint]</p> $m_{TR} = \frac{3}{4}$ $m_{SR} = -\frac{4}{3}$ $y = -\frac{4}{3}x + c$ $7 = -\frac{4}{3}(6) + c \quad \text{OR} \quad y - 7 = -\frac{4}{3}(x - 6)$ $y = -\frac{4}{3}x + 15 \quad y = -\frac{4}{3}x + 15$                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>✓ <math>x_R = 6</math> ✓ <math>y_R = 7</math></p> <p>✓ <math>m_{SR}</math></p> <p>✓ substitution of point R</p> <p>✓ equation</p> <p>(5)</p>                                                                                                                                                                                                                                           |
| 4.6 | <p>Let N be the centre of the circle.</p> $N\left(\frac{-3+2}{2}; \frac{3-1}{2}\right)$ $N\left(-\frac{1}{2}; 1\right)$ $AC = \sqrt{(2 - (-3))^2 + (3 - (-1))^2}$ $AC = \sqrt{41}$ $\left(x + \frac{1}{2}\right)^2 + (y - 1)^2 = \frac{41}{4}$ $x^2 + x + \frac{1}{4} + y^2 - 2y + 1 = \frac{41}{4}$ $x^2 + x + y^2 - 2y - 9 = 0$ $x^2 + x + y^2 - 2y - 9 = x^2 - 4x + y^2 - 8y - 5$ $6y + 5x - 4 = 0$ <p><b>OR</b></p> $\hat{B} = 90^\circ \quad [\angle \text{ in semi-circle} / \angle \text{ in halwe sirkel}]$ $\frac{y+1}{x+3} \times \frac{y-3}{x-2} = -1$ $y^2 - 2y - 3 = -1(x^2 + x - 6)$ $x^2 + x + y^2 - 2y - 9 = 0$ $x^2 + x + y^2 - 2y - 9 = x^2 - 4x + y^2 - 8y - 5$ $6y + 5x - 4 = 0$ | <p>✓ midpoint of AC: <math>N\left(-\frac{1}{2}; 1\right)</math></p> <p>✓ substitution into distance formula</p> <p>✓ length of AC</p> <p>✓ equation of new circle</p> <p>✓ equating</p> <p>✓ answer</p> <p>(6)</p> <p>✓ <math>\frac{y+1}{x+3} \times \frac{y-3}{x-2} = -1</math></p> <p>✓ simplification</p> <p>✓ equation of new circle</p> <p>✓ equating</p> <p>✓ answer</p> <p>(6)</p> |
| 4.7 | <p>The radius of the circle gets smaller. /<br/>Die radius van die sirkel word kleiner.</p> <p><b>OR</b></p> <p>The circle will decrease in size. /<br/>Die sirkel word kleiner.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>✓ correct statement</p> <p>(1)</p> <p>✓ correct statement</p> <p>(1)</p>                                                                                                                                                                                                                                                                                                               |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>[22]</b>                                                                                                                                                                                                                                                                                                                                                                               |

**QUESTION/VRAAG 5**

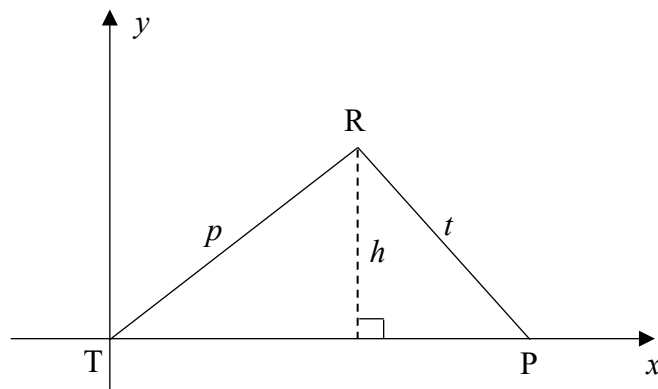
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.1.1 | <p>Given: <math>\sin(A - B) = \sin A \cos B - \cos A \sin B</math></p> <p><math>\sin(A + B) = \sin(A - (-B))</math></p> <p><math>= \sin A \cos(-B) - \cos A \sin(-B)</math></p> <p><math>= \sin A(\cos B) - \cos A(-\sin B)</math></p> <p><math>= \sin A \cos B + \cos A \sin B</math></p>                                                                                                                                                                                                                                                                                                                                                  | <p>✓ <math>\sin(A - (-B))</math></p> <p>✓</p> <p><math>\sin A \cos(-B) - \cos A \sin(-B)</math></p> <p>(2)</p>                                                                                                                                        |
| 5.1.2 | <p><math>\sqrt{3} \cos x = \sin(50^\circ + x) \cdot \cos(10^\circ - x) + \cos(50^\circ + x) \cdot \sin(10^\circ - x)</math></p> <p><math>\sqrt{3} \cos x = \sin[(50^\circ + x) + (10^\circ - x)]</math></p> <p><math>\sqrt{3} \cos x = \sin 60^\circ</math></p> <p><math>\sqrt{3} \cos x = \frac{\sqrt{3}}{2}</math></p> <p><math>\cos x = \frac{1}{2}</math></p> <p>ref <math>\angle = 60^\circ</math></p> <p><math>x = 60^\circ</math> or <math>x = 300^\circ</math></p>                                                                                                                                                                  | <p>✓ <math>\sin 60^\circ</math></p> <p>✓ special angle value</p> <p>✓ answer ✓ answers</p> <p>(4)</p>                                                                                                                                                 |
| 5.2   | <p><math display="block">\text{LHS} = \frac{\sin 2x}{2 \cos^2 x} + \frac{\sin x}{\tan x [1 + \cos(90^\circ - x)]}</math></p> <p><math display="block">= \frac{2 \sin x \cos x}{2 \cos^2 x} + \frac{\sin x}{\frac{\sin x}{\cos x} (1 + \sin x)}</math></p> <p><math display="block">= \frac{\sin x}{\cos x} + \frac{\cos x}{1 + \sin x}</math></p> <p><math display="block">= \frac{\sin x + \sin^2 x + \cos^2 x}{\cos x (1 + \sin x)}</math></p> <p><math display="block">= \frac{\sin x + 1}{\cos x (1 + \sin x)}</math></p> <p><math display="block">= \frac{1}{\cos x}</math></p> <p><math>\therefore \text{LHS} = \text{RHS}</math></p> | <p>✓ <math>\sin 2x = 2 \sin x \cos x</math></p> <p>✓ <math>\tan x = \frac{\sin x}{\cos x}</math></p> <p>✓ <math>\cos(90^\circ - x) = \sin x</math></p> <p>✓ writing as a single fraction</p> <p>✓ <math>\sin^2 x + \cos^2 x = 1</math></p> <p>(5)</p> |
| 5.3   | <p><math>\text{LHS} = \sin 260^\circ</math></p> <p><math>= -\sin 80^\circ</math></p> <p><math>= -\cos 10^\circ</math></p> <p><math>\text{RHS} = -\sqrt{\frac{\cos 20^\circ + 1}{2}}</math></p> <p><math>= -\sqrt{\frac{(2 \cos^2 10^\circ - 1) + 1}{2}}</math></p> <p><math>= -\sqrt{\cos^2 10^\circ}</math></p> <p><math>= -\cos 10^\circ</math></p> <p><math>\therefore \text{LHS} = \text{RHS}</math></p>                                                                                                                                                                                                                                | <p>✓ <math>\sin 260^\circ = -\sin 80^\circ</math></p> <p>✓ <math>\sin 80^\circ = \cos 10^\circ</math></p> <p>✓ <math>\cos 20^\circ = 2 \cos^2 10^\circ - 1</math></p> <p>✓ <math>-\sqrt{\cos^2 10^\circ}</math></p> <p>(4)</p>                        |
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>[15]</b>                                                                                                                                                                                                                                           |

**QUESTION/VRAAG 6**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.1.1 | $\begin{aligned} & \cos 330^\circ \cdot \cos 365^\circ + \sin 175^\circ \cdot \sin 210^\circ \\ &= \cos 30^\circ \cdot \cos 5^\circ + \sin 5^\circ (-\sin 30^\circ) \\ &= \cos 30^\circ \cdot \cos 5^\circ - \sin 30^\circ \cdot \sin 5^\circ \\ &= \cos(30^\circ + 5^\circ) \\ &= \cos 35^\circ \end{aligned}$ <p><b>OR</b></p> $\begin{aligned} & \cos 330^\circ \cdot \cos 365^\circ + \sin 175^\circ \cdot \sin 210^\circ \\ &= \sin 60^\circ \cdot \cos 5^\circ + \sin 5^\circ (-\cos 60^\circ) \\ &= \sin 60^\circ \cdot \cos 5^\circ - \sin 5^\circ \cdot \cos 60^\circ \\ &= \sin(60^\circ - 5^\circ) \\ &= \sin 55^\circ \end{aligned}$ | $\checkmark \cos 330^\circ = \cos 30^\circ$<br>$\checkmark \cos 365^\circ = \cos 5^\circ$<br>$\checkmark \sin 175^\circ = \sin 5^\circ$<br>$\checkmark \sin 210^\circ = -\sin 30^\circ$<br>$\checkmark$ answer (5)<br><br>$\checkmark \cos 330^\circ = \sin 60^\circ$<br>$\checkmark \cos 365^\circ = \cos 5^\circ$<br>$\checkmark \sin 175^\circ = \sin 5^\circ$<br>$\checkmark \sin 210^\circ = -\cos 60^\circ$<br>$\checkmark$ answer (5) |
| 6.1.2 | $\begin{aligned} & \cos 35^\circ = p \\ & \sin 70^\circ \\ &= 2 \sin 35^\circ \cos 35^\circ \\ &= 2p\sqrt{1-p^2} \end{aligned}$  <p><b>OR</b></p>  $\begin{aligned} & \sin 70^\circ \\ &= \cos 20^\circ \\ &= \cos(55^\circ - 35^\circ) \\ &= \cos 55^\circ \cdot \cos 35^\circ + \sin 55^\circ \cdot \sin 35^\circ \\ &= \sqrt{1-p^2} \cdot p + p\sqrt{1-p^2} \\ &= 2p\sqrt{1-p^2} \end{aligned}$                                                                        | $\checkmark$ 3rd side = $\sqrt{1-p^2}$<br>$\checkmark$ expansion of double $\angle$<br>$\checkmark$ substitution (3)<br><br>$\checkmark$ 3rd side = $\sqrt{1-p^2}$<br>$\checkmark$ expansion of compound $\angle$<br>$\checkmark$ substitution (3)                                                                                                                                                                                           |

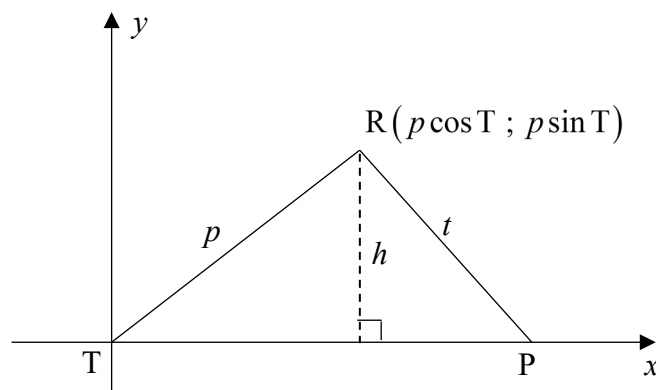
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.2         | $\frac{2^{3\sin^2 x}}{4 \cdot 2^{\sin x}} = 1$ $\frac{2^{3\sin^2 x}}{2^{\sin x + 2}} = 1$ $2^{3\sin^2 x} = 2^{\sin x + 2}$ $3\sin^2 x = \sin x + 2$ $3\sin^2 x - \sin x - 2 = 0$ $(3\sin x + 2)(\sin x - 1) = 0$ $\sin x = -\frac{2}{3} \quad \text{or} \quad \sin x = 1 \text{ (N/A)}$ $\text{ref } \angle = 41,810\dots^\circ$ $x = 221,81^\circ + k \cdot 360^\circ \text{ or } x = 318,19^\circ + k \cdot 360^\circ, k \in \mathbb{Z}$ <p><b>OR</b></p> $\frac{2^{3\sin^2 x}}{4 \cdot 2^{\sin x}} = 1$ $\frac{2^{3\sin^2 x}}{2^2 \cdot 2^{\sin x}} = 2^0$ $2^{3\sin^2 x - 2 - \sin x} = 2^0$ $3\sin^2 x - \sin x - 2 = 0$ $(3\sin x + 2)(\sin x - 1) = 0$ $\sin x = -\frac{2}{3} \quad \text{or} \quad \sin x = 1 \text{ (N/A)}$ $\text{ref } \angle = 41,810\dots^\circ$ $x = 221,81^\circ + k \cdot 360^\circ \text{ or } x = 318,19^\circ + k \cdot 360^\circ, k \in \mathbb{Z}$ | <p>✓ exponential law</p> <p>✓ cross multiplication</p> <p>✓ equation from exponents</p> <p>✓ equation</p> <p>✓ both <math>x</math> - values<br/>✓ <math>+k \cdot 360^\circ; k \in \mathbb{Z}</math></p> <p>(6)</p> <p>✓ <math>2^0</math></p> <p>✓ exponential law</p> <p>✓ equation from exponents</p> <p>✓ equation</p> <p>✓ both <math>x</math> - values<br/>✓ <math>+k \cdot 360^\circ; k \in \mathbb{Z}</math></p> <p>(6)</p> |
| <b>[14]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                   |



**QUESTION/VRAAG 8**

|     |                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.1 | <p>Construction of perpendicular height, <math>h</math></p> $\sin T = \frac{h}{p} \quad \text{and} \quad \sin P = \frac{h}{t}$ $h = p \sin T \quad \text{and} \quad h = t \sin P$ $p \sin T = t \sin P$ $\frac{p \sin T}{pt} = \frac{t \sin P}{pt}$ $\frac{\sin T}{t} = \frac{\sin P}{p}$ | <p>✓ correct construction</p> <p>✓ <math>\sin T = \frac{h}{p} \quad \text{and} \quad \sin P = \frac{h}{t}</math></p> <p>✓ <math>h = p \sin T \quad \text{and} \quad h = t \sin P</math></p> <p>✓ <math>p \sin T = t \sin P</math></p> |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

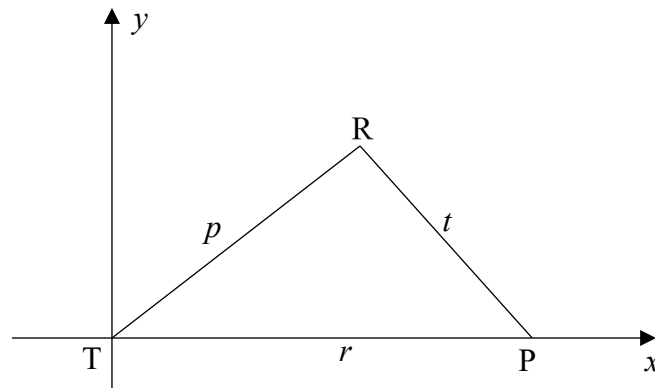
(4)

**OR**

|     |                                                                                                                                                                       |                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.1 | <p><math>R(p \cos T ; p \sin T)</math></p> $h = p \sin T$ $\sin P = \frac{h}{t}$ $\sin P = \frac{p \sin T}{t}$ $\frac{\sin T}{t} = \frac{\sin P}{p}$ <p><b>OR</b></p> | <p>✓ <math>x_R = p \cos T \quad \checkmark \quad y_R = p \sin T</math></p> <p>✓ <math>\sin P = \frac{h}{t}</math></p> <p>✓ <math>\sin P = \frac{p \sin T}{t}</math></p> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(4)

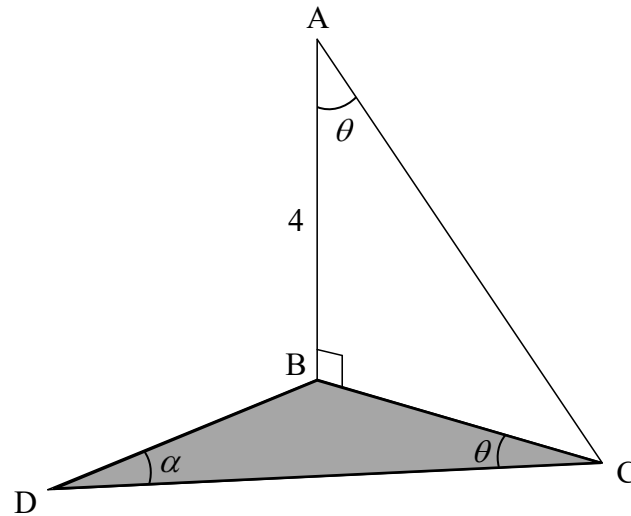
8.1



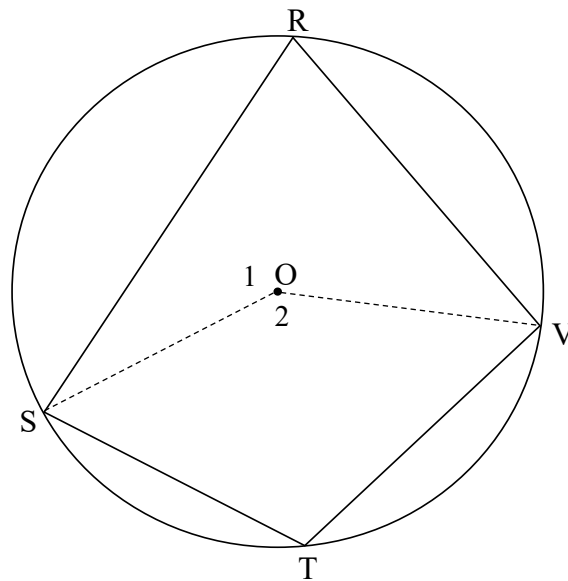
|  |                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | $\text{Area of } \triangle PRT = \frac{1}{2} pr \sin T$ $\text{Area of } \triangle PRT = \frac{1}{2} tr \sin P$ $\therefore \frac{1}{2} pr \sin T = \frac{1}{2} tr \sin P$ $\frac{\frac{1}{2} pr \sin T}{\frac{1}{2} ptr} = \frac{\frac{1}{2} tr \sin P}{\frac{1}{2} ptr}$ $\frac{\sin T}{t} = \frac{\sin P}{p}$ | $\checkmark \text{ Area of } \triangle PRT = \frac{1}{2} pr \sin T$ $\checkmark \text{ Area of } \triangle PRT = \frac{1}{2} tr \sin P$ $\checkmark \text{ equating of areas}$ $\checkmark \text{ dividing by } \frac{1}{2} ptr$ |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(4)

8.2

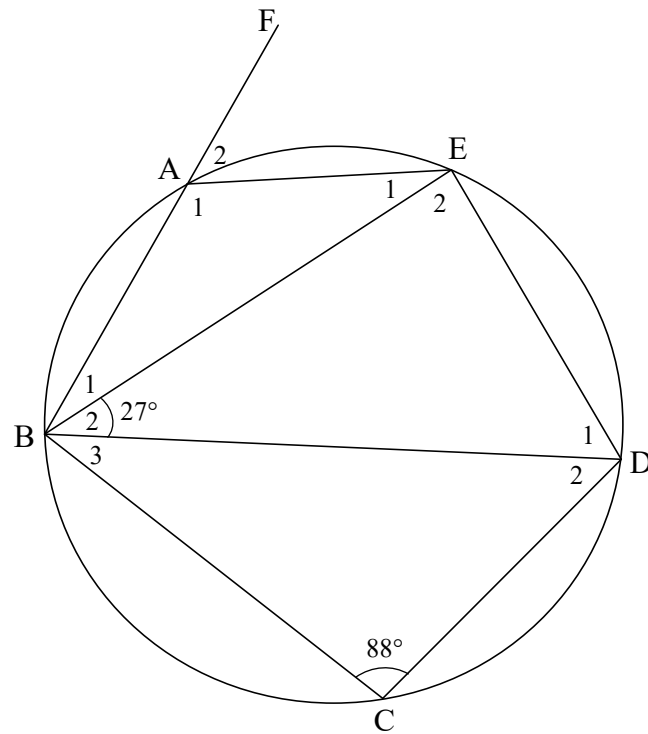


|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                               |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.2.1       | $\tan \theta = \frac{BC}{4}$ $BC = 4 \tan \theta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\checkmark \tan \theta = \frac{BC}{4}$ <p style="text-align: right;">(1)</p>                                                                                                                                                                                                                 |
| 8.2.2       | $\hat{D}BC = 180^\circ - (\alpha + \theta)$ $\frac{DC}{\sin[180^\circ - (\alpha + \theta)]} = \frac{4 \tan \theta}{\sin \alpha}$ $DC = \frac{4 \tan \theta \sin(\alpha + \theta)}{\sin \alpha}$ $= \frac{4 \tan \theta (\sin \alpha \cdot \cos \theta + \cos \alpha \cdot \sin \theta)}{\sin \alpha}$ $= \frac{4 \sin \theta (\sin \alpha \cdot \cos \theta + \cos \alpha \cdot \sin \theta)}{\cos \theta \cdot \sin \alpha}$ $= 4 \sin \theta \left( \frac{\sin \alpha \cdot \cos \theta}{\cos \theta \cdot \sin \alpha} + \frac{\cos \alpha \cdot \sin \theta}{\cos \theta \cdot \sin \alpha} \right)$ $= 4 \sin \theta \left( 1 + \tan \theta \frac{1}{\tan \alpha} \right)$ $= 4 \sin \theta \left( 1 + \frac{\tan \theta}{\tan \alpha} \right)$ | $\checkmark$ correct use of sine rule<br>$\checkmark \sin[180^\circ - (\alpha + \theta)] = \sin(\alpha + \theta)$<br>$\checkmark$ compound angle expansion<br><br>$\checkmark$ separating into two fractions<br>$\checkmark$ using quotient identity<br><p style="text-align: right;">(5)</p> |
| <b>[10]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                               |

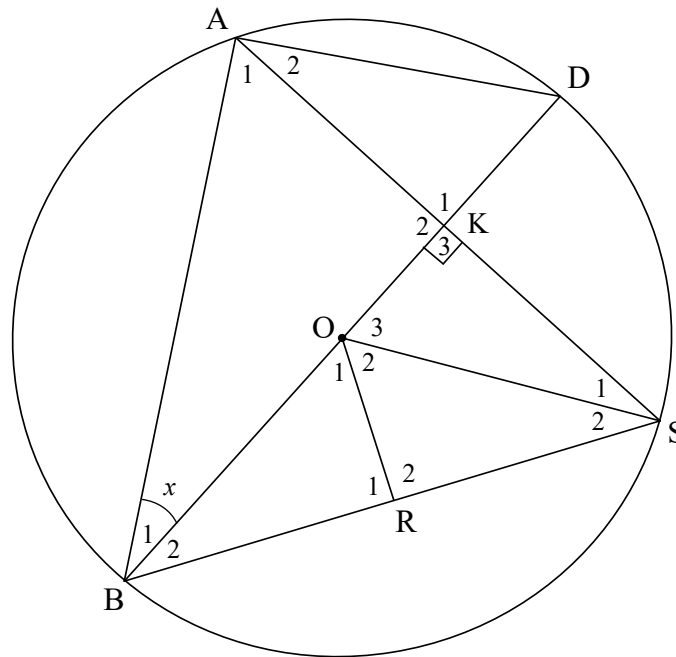
**QUESTION/VRAAG 9**

|     |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                     |                                                              |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 9.1 | <p>Draw radii OS and OV</p> <p><math>\hat{O}_2 = 2\hat{R}</math></p> <p><math>\hat{O}_1 = 2\hat{T}</math></p> <p><math>\hat{O}_1 + \hat{O}_2 = 360^\circ</math></p> <p><math>2\hat{R} + 2\hat{T} = 360^\circ</math></p> <p><math>2(\hat{R} + \hat{T}) = 360^\circ</math></p> <p><math>\hat{R} + \hat{T} = 180^\circ</math></p> | <p>[<math>\angle</math> at centre = <math>2 \times \angle</math> at circumference/<br/>midpts <math>\angle = 2 \times</math> omtreks <math>\angle</math>]</p> <p>[<math>\angle</math> at centre = <math>2 \times \angle</math> at circumference<br/>midpts <math>\angle = 2 \times</math> omtreks <math>\angle</math>]</p> <p>[<math>\angle</math>s round a pt/<math>\angle</math>e om 'n punt]</p> | <p>✓ construction</p> <p>✓ S ✓ R</p> <p>✓ S</p> <p>✓ S/R</p> |
| (5) |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                     |                                                              |

## 9.2



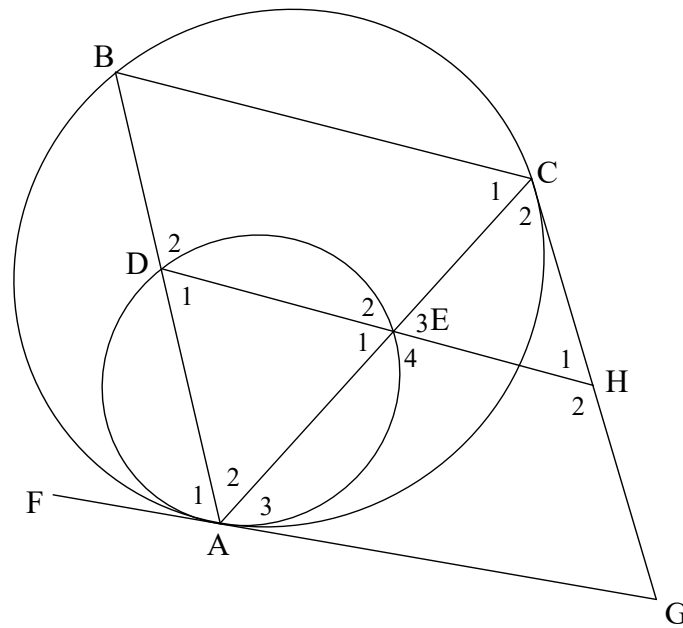
|       |                                                             |                                                                                                                       |                |      |
|-------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------|------|
| 9.2.1 | $\hat{E}_2 = 92^\circ$                                      | [opp $\angle$ s of cyclic quad/teenoorst $\angle$ e v kvh]                                                            | ✓ S ✓ R        | (2)  |
| 9.2.2 | $\hat{B}_3 = \hat{B}_2 = 27^\circ$                          | [equal chords ; equal $\angle$ s/<br>gelyke koorde; gelyke $\angle$ e]                                                | ✓ S ✓ R        | (2)  |
| 9.2.3 | $\hat{D}_1 = 61^\circ$<br>$\therefore \hat{A}_2 = 61^\circ$ | [sum of $\angle$ s in a $\Delta$ /binne $\angle$ e v $\Delta$ ]<br>[ext $\angle$ of cyclic quad/buite $\angle$ v kvh] | ✓ S<br>✓ S ✓ R | (3)  |
|       |                                                             |                                                                                                                       |                | [12] |

**QUESTION/VRAAG 10**

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------|
| 10.1   | <p>The angles subtended by a chord at the circumference of the circle, on the same side of the chord, are <b>equal</b>.</p> <p><i>Die hoeke wat deur 'n koord op die sirkel aan dieselfde kant van die koord onderspan word, is <b>gelyk</b>.</i></p>                                                                                                                                                                                                                                                                                                                  | ✓ S                                  | (1)                |
| 10.2.1 | <p><math>\hat{DAB} = 90^\circ</math> [<math>\angle</math> in semi-circle/<math>\angle</math> in halwe sirkel]</p> <p><math>\hat{D} = 90^\circ - x</math> [sum of <math>\angle</math>s in a <math>\Delta</math>/binne <math>\angle</math>e v <math>\Delta</math>]</p>                                                                                                                                                                                                                                                                                                   | ✓ S ✓ R<br>✓ S/R                     | (3)                |
| 10.2.2 | <p><math>\hat{S}_1 + \hat{S}_2 = 90^\circ - x</math> [<math>\angle</math>s in the same segment/<math>\angle</math>e in dies. segment]</p> <p><math>\hat{B}_2 = x</math> [sum of <math>\angle</math>s in a <math>\Delta</math>/binne <math>\angle</math>e v <math>\Delta</math>]</p> <p><b>OR</b></p> <p><math>\hat{A}_2 = x</math> [ext <math>\angle</math> of <math>\Delta</math>/ buite <math>\angle</math> v <math>\Delta</math>]</p> <p><math>\therefore \hat{B}_2 = x</math> [<math>\angle</math>s in the same segment/<math>\angle</math>e in dies. segment]</p> | ✓ S ✓ R<br>✓ S<br><br>✓ S<br>✓ S ✓ R | (3)<br><br><br>(3) |
| 10.2.3 | <p><math>\hat{S}_2 = x</math> [<math>\angle</math>s opp equal sides/<math>\angle</math>e teenoor gelyke sye]</p> <p><math>\hat{S}_1 = 90^\circ - 2x</math></p> <p><b>OR</b></p> <p><math>\hat{O}_3 = 2x</math> [<math>\angle</math> at centre = <math>2 \times \angle</math> at circumference<br/>midpts <math>\angle = 2 \times</math> omtreks <math>\angle</math>]</p> <p><math>\hat{S}_1 = 90^\circ - 2x</math> [sum of <math>\angle</math>s in a <math>\Delta</math>/ binne <math>\angle</math>e v <math>\Delta</math>]</p> <p><b>OR</b></p>                       | ✓ S/R<br>✓ S<br><br>✓ S/R<br>✓ S     | (2)<br><br><br>(2) |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|             | $\hat{D} = \hat{S}_1 + \hat{S}_2$ [∠s in the same segment/∠e in dies. segment]<br>$90^\circ - x = \hat{S}_1 + x$<br>$\therefore \hat{S}_1 = 90^\circ - 2x$ [sum of ∠s in a Δ/ binne ∠e v Δ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ✓ S/R<br><br>✓ S<br>(2)                                                                     |
| 10.3        | $\hat{R}_1 = \hat{R}_2 = 90^\circ$ [line from centre to midpt of chord/<br>lyn vanuit midpt na midpt van koord]<br>$\therefore \hat{R}_2 + \hat{K}_3 = 180^\circ$<br>$\therefore$ OKSR is a cyclic quadrilateral [converse opp ∠s of cyclic quad/<br>omgekeerde teenoorst. ∠e v kvh]<br><b>OR</b><br>$\hat{K}_2 = 90^\circ$ [∠s on a str line/∠e op 'n reguitlyn]<br>$\hat{R}_1 = \hat{R}_2 = 90^\circ$ [line from centre to midpt of chord/<br>lyn vanuit midpt na midpt van koord]<br>$\therefore \hat{R}_2 = \hat{K}_2 = 90^\circ$<br>$\therefore$ OKSR is a cyclic quadrilateral [converse ext ∠ of cyclic quad/<br>omgekeerde buite ∠ v kvh]<br><b>OR</b><br>$\hat{A}\hat{D}\hat{K} = \hat{K}\hat{S}\hat{R} = 90^\circ - x$ [∠s in the same segment/<br>∠e in dies. segment]<br>$\hat{R}_1 = \hat{R}_2 = 90^\circ$ [line from centre to midpt of chord/<br>lyn vanuit midpt na midpt van koord]<br>$\hat{B}\hat{O}\hat{R} = 90^\circ - x$ [sum of ∠s in a Δ/ binne ∠e v Δ]<br>$\therefore \hat{B}\hat{O}\hat{R} = \hat{K}\hat{S}\hat{R} = 90^\circ - x$<br>$\therefore$ OKSR is a cyclic quadrilateral [converse ext ∠ of cyclic quad/<br>omgekeerde buite ∠ v kvh] | ✓ S ✓ R<br><br>✓ R<br>(3)<br><br>✓ S ✓ R<br><br>✓ R<br>(3)<br><br>✓ S ✓ R<br><br>✓ R<br>(3) |
| <b>[12]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                             |

## QUESTION/VRAAG 11



|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 11.1 | <p>In <math>\triangle ADE</math> and <math>\triangle ABC</math></p> <p>1. <math>\hat{A}_2 = \hat{A}_2</math> [common/gemeenskaplik]</p> <p>2. <math>\hat{A}_1 = \hat{E}_1</math> [tan chord theorem/raaklyn koord st]<br/> <math>\hat{A}_1 = \hat{C}_1</math> [tan chord theorem/ raaklyn koord st]<br/> <math>\therefore \hat{E}_1 = \hat{C}_1</math></p> <p>3. <math>\hat{A}_3 = \hat{D}_1</math> [tan chord theorem/ raaklyn koord st]<br/> <math>\hat{A}_3 = \hat{B}</math> [tan chord theorem/ raaklyn koord st]<br/> <math>\therefore \hat{D}_1 = \hat{B}</math></p> <p><math>\therefore \triangle ADE \parallel \triangle ABC</math> [<math>\angle \angle \angle</math>]</p>                                                                                                                                                  | <p>✓ S</p> <p>✓ S ✓ R</p> <p>✓ S OR R</p> <p>(4)</p>                                                  |
| 11.2 | <p><math>\hat{D}_1 = \hat{B}</math></p> <p><math>DE \parallel BC</math> [corresp <math>\angle</math>s =/ooreenk. <math>\angle</math>e = ]</p> <p><math>\frac{AD}{DB} = \frac{AE}{EC}</math> [prop theorem; <math>DE \parallel BC</math>/eweredige st.; <math>DE \parallel BC</math>]<br/> <b>OR</b><br/> [line <math>\parallel</math> one side of <math>\triangle</math>/lyn <math>\parallel</math> een sy v <math>\triangle</math> ]</p> <p><math>\frac{1k}{2k} = \frac{4}{EC}</math><br/> <math>\therefore EC = 8</math></p> <p><b>OR</b></p> <p><math>\frac{AD}{AB} = \frac{AE}{AC}</math> [<math>\triangle ADE \parallel \triangle ABC</math>]<br/> <math>\frac{1k}{3k} = \frac{4}{AC}</math> [<math>AB = 3AD</math>]<br/> <math>AC = 12</math><br/> <math>\therefore EC = 12 - 4</math><br/> <math>\therefore EC = 8</math></p> | <p>✓ S/R</p> <p>✓ S ✓ R</p> <p>✓ answer</p> <p>(4)</p> <p>✓ S ✓ R</p> <p>✓ AC</p> <p>✓ answer (4)</p> |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 11.3 | <p>In <math>\triangle ECH</math> and <math>\triangle CBA</math></p> <p><math>\hat{E}_3 = \hat{C}_1</math> [alt <math>\angle</math>s; <math>BC \parallel DE</math>/verw. <math>\angle</math>e; <math>BC \parallel DE</math>]</p> <p><math>\hat{C}_2 = \hat{B}</math> [tan chord theorem/raaklyn koord st.]</p> <p><math>\hat{H}_1 = \hat{A}_2</math> [sum of <math>\angle</math>s in a <math>\triangle</math> / binne <math>\angle</math>e v <math>\triangle</math>]</p> <p><math>\therefore \triangle ECH \parallel \triangle CBA</math> [<math>\angle \angle \angle</math>]</p> <p><math>\therefore \frac{EH}{EC} = \frac{CA}{CB}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>✓ S/R</p> <p>✓ S ✓ R</p> <p>✓ S <b>OR</b> R</p> <p>(4)</p>                                              |
| 11.4 | <p><math>\frac{EH}{EC} = \frac{CA}{CB}</math> [ proved in 11.3]</p> <p><math>EC = \frac{EH \cdot BC}{AC}</math></p> <p><math>\frac{EC}{AC} = \frac{BD}{AB}</math> [prop theorem; <math>DE \parallel BC</math>/eweredige st.; <math>DE \parallel BC</math>]<br/><b>OR</b><br/>[line <math>\parallel</math> one side of <math>\triangle</math>/lyn <math>\parallel</math> een sy v <math>\triangle</math>]</p> <p><math>EC = \frac{AC \cdot BD}{AB}</math></p> <p><math>EC \times EC = \frac{EH \cdot BC}{AC} \times \frac{AC \cdot BD}{AB}</math></p> <p><math>EC^2 = \frac{EH \cdot BC \cdot DB}{AB}</math></p> <p><math>EC^2 = \frac{EH \cdot BC \cdot DB}{3AD}</math> [AD: AB = 1:3]</p> <p><math>EC = \sqrt{\frac{EH \cdot BC \cdot DB}{3AD}}</math></p> <p><b>OR</b></p> <p><math>\frac{EH}{EC} = \frac{CA}{CB}</math> [ proved in 11.3]</p> <p><math>EC = \frac{EH \cdot BC}{AC}</math></p> <p><math>\frac{EC}{AC} = \frac{BD}{AB}</math> [prop theorem; <math>DE \parallel BC</math>/eweredige st.; <math>DE \parallel BC</math>]<br/><b>OR</b><br/>[line <math>\parallel</math> one side of <math>\triangle</math>/lyn <math>\parallel</math> een sy v <math>\triangle</math>]</p> <p><math>AC = \frac{AB \cdot EC}{BD}</math></p> | <p>✓ S</p> <p>✓ S / R</p> <p>✓ S</p> <p>✓ S</p> <p>✓ S</p> <p>(5)</p> <p>✓ S</p> <p>✓ S / R</p> <p>✓ S</p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                              |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | $EC = \frac{EH \cdot BC}{AB \cdot EC}$ $EC = \frac{EH \cdot BC \cdot DB}{AB \cdot EC}$ $EC^2 = \frac{EH \cdot BC \cdot DB}{3AD} \quad [AD: AB = 1:3]$ $EC = \sqrt{\frac{EH \cdot BC \cdot DB}{3AD}}$ <p><b>OR</b></p> $\frac{AD}{AB} = \frac{AE}{AC} \quad [\triangle ADE \parallel \triangle ABC]$ $AB = \frac{AD \cdot AC}{AE}$ $3AD = \frac{AD \cdot AC}{AE} \quad [AD: AB = 1:3]$ $3 = \frac{AC}{AE}$ $\frac{EH}{EC} = \frac{CA}{CB} \quad [\text{proved in 11.3}]$ $AC = \frac{EH \cdot BC}{EC}$ $\frac{AD}{DB} = \frac{AE}{EC} \quad [\text{prop theorem; } DE \parallel BC / \text{eweredige st.; } DE \parallel BC]$ <p style="text-align: center;"><b>OR</b></p> $[line \parallel \text{one side of } \triangle / \text{lyn} \parallel \text{een sy v } \triangle]$ $AE = \frac{AD \cdot EC}{DB}$ $\therefore 3 = \frac{EH \cdot BC \cdot DB}{EC \cdot AD \cdot EC}$ $3 EC^2 = \frac{EH \cdot BC \cdot DB}{AD}$ $EC = \sqrt{\frac{EH \cdot BC \cdot DB}{3AD}}$ | <p style="text-align: right;">(5)</p> <p>✓ S</p> <p>✓ S</p> <p>✓ S</p> <p>✓ S</p> <p>✓ S</p> <p>✓ S / R</p> <p>✓ S</p> <p style="text-align: right;">(5)</p> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [17]                                                                                                                                                         |

**TOTAL/TOTAAL: 150**